

Insurance & Quantification

A Sentiment-Enhanced Volatility Prediction Project

GitHub: <https://github.com/ffgh-code/insuranceandquantification>

Live Demo: <https://insuranceandquantification-fbmyvwetabtki2r5m8krac.streamlit.app/>

1. Why This Project

I am an actuarial science student transitioning into quantitative finance. Traditional volatility models like GARCH only use historical price data, but market movements are heavily driven by sentiment - fear, greed, policy signals, and geopolitical risks. I wanted to test whether quantifying news sentiment as a numerical signal and feeding it into volatility models could improve prediction accuracy, especially for the Chinese A-share market where policy sensitivity is high.

2. Data

Market data: CSI 300 index (000300.SH) via akshare, approximately 727 trading days of real daily OHLCV data from the Chinese A-share market. Sentiment data: 20 Chinese financial news headlines covering interest rate cuts, exchange rates, real estate, AI, employment, and geopolitics. The system also supports US market data via yfinance and falls back to realistic synthetic data when APIs are unavailable.

3. Methodology

Sentiment Analysis: Two approaches compared side by side. LLM-powered analysis using OpenAI-compatible APIs with a structured prompt returning directional scores and topic classification. Traditional VADER lexicon-based analysis as a baseline. When the LLM API is unavailable, a financial keyword lexicon with 40+ weighted positive and 30+ weighted negative terms serves as a rule-based fallback.

Volatility Modeling: Four realized volatility estimators implemented - Close-to-Close, Parkinson, Garman-Klass, and Yang-Zhang. GARCH family models fitted including standard GARCH(1,1), EGARCH for asymmetric volatility, and GJR-GARCH. An LSTM neural network with 2 hidden layers and 64 units per layer provides a deep learning comparison.

Strategy Backtesting: Four trading strategies compared - volatility mean reversion, pure sentiment-driven, combined (both signals agree), and volatility risk premium. The backtesting engine accounts for transaction costs (0.1% per trade), slippage (0.05%), and reports full risk metrics including Sharpe ratio, Sortino ratio, Calmar ratio, and maximum drawdown.

4. Actuarial Extensions

Three additional modules bridge the quant finance framework with actuarial science: Solvency II / C-ROSS capital requirement calculation maps GARCH volatility forecasts to regulatory solvency capital. Loss reserving applies GARCH adjustments to traditional reserving methods. Mortality forecasting compares Lee-Carter, GARCH, and naive methods for mortality improvement rate prediction.

5. Results

Real CSI 300 data: 727 trading days of real Chinese market data. GARCH(1,1): AIC of 2,030, indicating a valid model fit on Chinese index returns. Best strategy: Combined sentiment-volatility approach with a Sharpe ratio of 0.43. The combined strategy outperformed pure volatility mean reversion and pure sentiment-driven approaches individually.

Key observation: The Sharpe ratio on real A-share data (0.43) is lower than on synthetic US data (1.03), reflecting the different volatility structure of Chinese markets - stronger trending behavior and weaker mean reversion. This result itself is an interesting empirical finding worth further investigation.

6. What I Learned

This project taught me how to bridge traditional econometric models with modern machine learning techniques in a practical quant finance setting. I gained hands-on experience with time series analysis, GARCH modeling, LSTM neural networks, LLM-based NLP, and strategy backtesting. The actuarial extensions demonstrated how the same volatility modeling framework applies across both finance and insurance domains. The modular design of the project makes it easy to extend with new data sources, models, and strategies.

Links

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Medium Article: <https://medium.com/@ffgh-code> (link pending)